

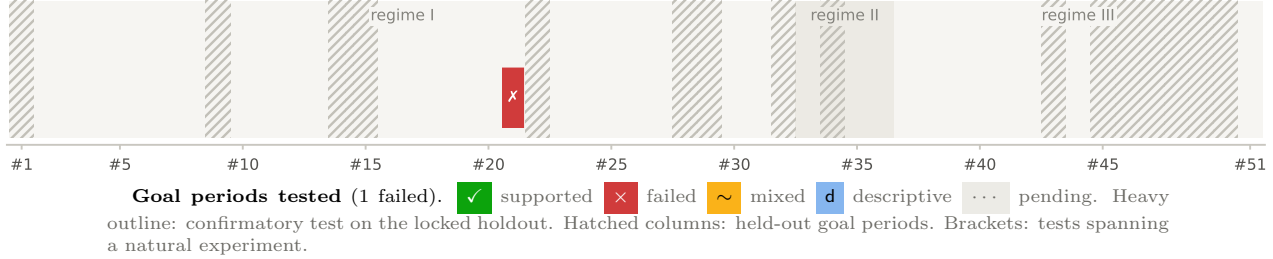
H24: In the forecast week (#21), switching from independent drafts to comparison switches the coupling on

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-03

Complete 40% Faithfulness 1/5 Usefulness 1.5/5

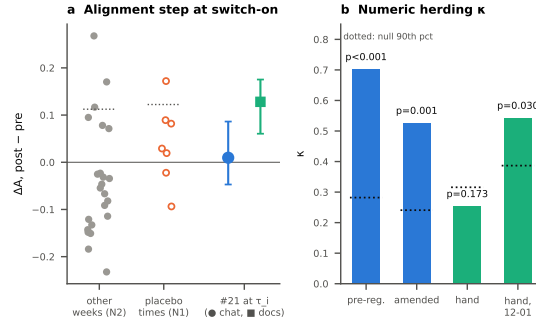
Bottom line. In the forecast week the coupling switch was real (teammate mentions jumped from 7% to 74%) but early, and it moved neither content alignment nor, after an audit, the numeric forecasts: the agents were aligned from the first hour.

level: hypothesis · pipeline checks 86% · updated 2026-10-03



WHAT HAPPENED

Goal period #21 (forecast AI; regime I; 8–9 agents) as a natural experiment. The switch-on was located from the record before any outcome. Agents’ session intentions and their own screen descriptions put each agent’s first reading of a teammate’s forecast 42–86 min into the week (median 50 min), not mid-week. The share of messages naming a teammate jumps there from 7% to 74%. The predicted alignment step did not appear: $\Delta A_{\text{res}} = +0.010 [-0.05, +0.09]$, inside both a within-week placebo and 23 other kickoff days. The synthetic gives ≈ 0.7 power at $\beta J_0/n \approx 0.55$. Content was already aligned before any reading ($A_{\text{res}} \approx 0.33$ vs. a rotation null of 0.03) and ramped over the week ($\rho = 0.73$). Numeric herding passed as pre-registered ($\kappa = 0.70$, $p < 0.001$) but was an extraction artifact: hand-verified values give $\kappa = 0.25$ ($p = 0.17$). Holdout (#14) not run.



Observables. (a) Change in residual alignment across the switch-on for #21 (blue: statements; green: documents; 90% CIs), against 23 other regime-I kickoff days at the same offset (gray) and placebo switch times inside #21 (orange). Dotted lines: 90th percentiles. (b) DeGroot pull κ under the pre-registered extractor, the amended extractor and hand-verified values; dotted: the independent-updating null, 90th percentile. The herding signal shrinks to noise once extraction errors are removed.

PHYSICAL MODEL

Model 11 (vector spins), with model 01’s Curie–Weiss analogue. Spin $\mathbf{s}_i$: agent i ’s whitened, unit-normalized statement embedding ($n = 32$). Field $h\hat{g}$: the goal and kickoff text. $\mathbf{h}_i$: agent style. Coupling J_0 : a mean-field pull toward the others, switched on at τ_i (the agent’s first reading of a teammate’s forecast). A : mean pairwise cosine, with $\hat{g}$ removed for A_{res} . Numeric forecasts x_i (AGI by 2035, SI by 2050, p(doom)) are soft spins with a DeGroot pull κ .

$$\mathbf{s}_i(t) \propto h\hat{g} + \mathbf{h}_i + \xi_i(t) + J_0 \Theta(t - \tau_i) \mathbf{m}_{-i}(t)$$

$$A = \frac{2}{N(N-1)} \sum_{i < j} \hat{\mathbf{v}}_i \cdot \hat{\mathbf{v}}_j, \quad \beta J_0/n = 1 - \frac{1}{1 + (N-1)A_{\text{res}}}$$

$$\Delta x_i = \kappa (\bar{x}_{-i} - x_i) + \epsilon_i, \quad \kappa = 0 \text{ for independent updating}$$

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

Mostly a cautionary tale. Telling LLM agents to “form independent views, then compare” did not buy independence: their content was aligned before anyone read anyone, from the shared prompt, instructions and prior. Comparison then produced talk about divergence, and some literal copying of numbers, not a measurable alignment jump. For anyone aggregating swarm forecasts, an independent-first-draft phase is no diversity guarantee: most of the diversity worth protecting is gone at kickoff. The reusable lessons are methodological. Regex-extracted numbers manufacture herding through regression to the mean, and snapshot coupling estimates absorb unmodelled fields. One week, $N = 7$, no ground truth: a pilot.

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